

WellnessDx – MVP Build Scope (Finalized)

Objective

Build a focused, cost-efficient MVP delivering structured evidence ingestion, dual summary formats (patient + clinician), population-level treatment impact insights, and alerts for engagement — while avoiding unnecessary complexity.

Module 1 – Evidence Ingestion & Transformation (Estimated: 150–210 hours)

- Semantic Scholar: structured study metadata extraction (study type, population, outcomes, duration, effect direction).
- Atomic claim storage tied to citations.
- USDA: nutrient & curated bioactive mapping; supplement-to-food grounding.
- NIH ODS: studied dosage ranges, ULs, safety flags, interaction sections.
- Schema normalization and structured storage in Postgres.

Module 2 – Retrieval & Ranking Layer (Estimated: 60–90 hours)

- Vector search for semantic recall.
- Metadata filtering (disease, symptom, intervention).
- Deterministic ranking (study type, recency, sample size).
- Citation enforcement and contradiction flagging.

Module 3 – Dual Summary System (Estimated: 80–120 hours)

- Patient Summary: simplified explanation, symptom impact, food vs supplement clarity, applicability boundaries.
- Clinician Summary: study hierarchy, population definitions, outcome consistency, dosage ranges, limitations.
- Shared evidence backbone with structured synthesis templates.

Module 4 – Population-Level Treatment Impact View (Estimated: 40–60 hours)

- Structured grouping by symptom.
- Strong / Mixed / Limited evidence categorization.
- Descriptive framing only (no prescriptive logic).

Module 5 – Alerts (Insights Agent) (Estimated: 40–60 hours)

- Keyword-based monitoring (condition, symptom, ingredient).
- New evidence detection via Semantic Scholar.
- Basic notification system (email or in-app).

Estimated Total MVP Build Time: 370–540 hours

This estimate reflects a disciplined MVP build without biomarker logic, mechanism graphing, OpenFoodFacts integration, or advanced personalization. Those are explicitly deferred to post-MVP.